

0 4

A student investigated the effect of ethanol, hydrochloric acid and temperature on the loss of red pigment from beetroot cells.

During the procedure, the student:

- added 10 cm³ water into one test tube
- added 10 cm³ ethanol into a second test tube
- added 10 cm³ hydrochloric acid into a third test tube
- put the three tubes into a 25 °C water bath
- cut four cylinders of tissue from a beetroot
- put a cylinder into each tube and fitted bungs
- added 10 cm³ water into a fourth test tube and put this tube into a 70 °C water bath
- placed the fourth cylinder into this tube and fitted a bung
- later removed the cylinders from the tubes
- estimated the intensity of red pigment in each solution by eyesight.

0 4 . 1

Give **one** way in which the student could ensure the first three beetroot cylinders were kept at 25 °C throughout her experiment.

[1 mark]

0 4 . 2

Give **two** variables that the student did **not** control in her procedure.

[2 marks]

1

2

Question 4 continues on the next page

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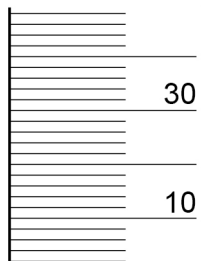


0 4 . 3

The student used a measuring cylinder to obtain 10 cm^3 of each solution.

Figure 2 shows some of the scale graduations on the side of this measuring cylinder.

Figure 2



What is the uncertainty of taking a reading of 10 cm^3 with this measuring cylinder?

Suggest how you could reduce the uncertainty calculated.

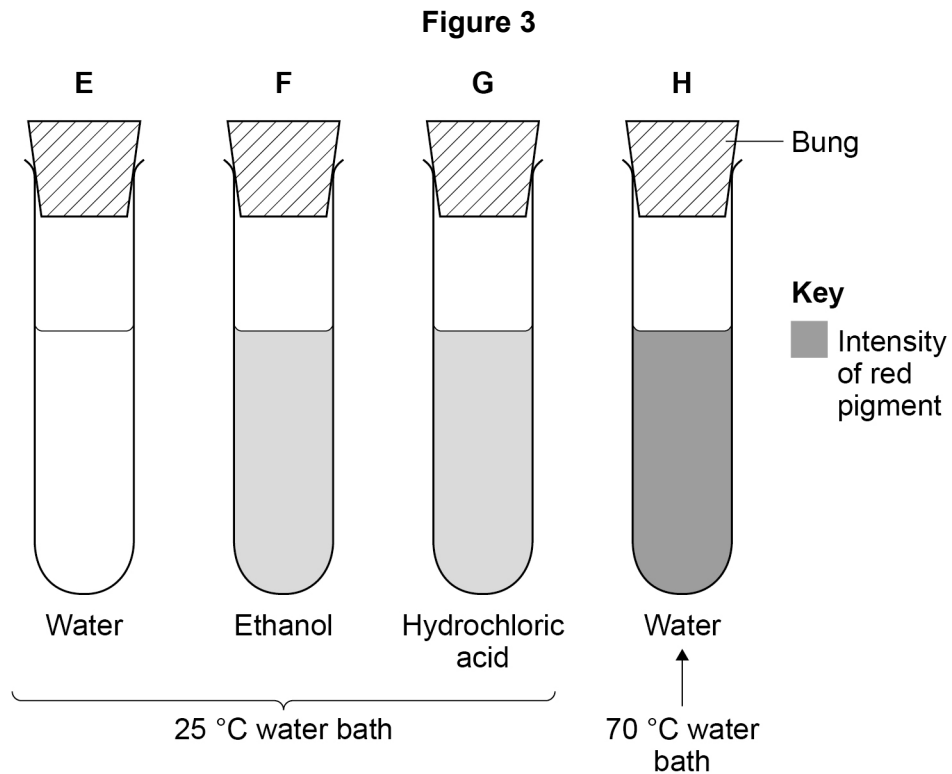
[2 marks]

Uncertainty $\pm$ _____ cm^3

Reducing uncertainty _____



A different student used the same procedure and she controlled **all** variables appropriately. Her results are shown in **Figure 3**.



0 4 . 4

Using **Figure 3**, what can you conclude about the damage caused to beetroot cells by water, ethanol, hydrochloric acid and different temperatures?

Provide explanations for your conclusions.

[4 marks]

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